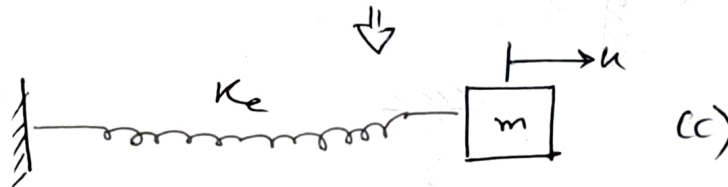
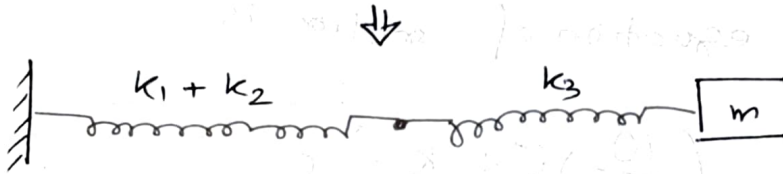


Homework 1
Structural Dynamics

1/17/2025

Q1

Solution:-



(a) $f_s = k_e u$
 $f_s = (k_1 + k_2) u$
 $k_e = f_s / u = k_1 + k_2$

(b) $\frac{f_s}{k_e} = \frac{f_s}{k_1} + \frac{f_s}{k_2} + \frac{f_s}{k_3}$

$\Rightarrow \frac{1}{k_e} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3}$

$\Rightarrow k_e = \frac{k_1 k_2}{k_1 + k_2}$

$$\frac{1}{k_e} = \frac{1}{k_1 + k_2} + \frac{1}{k_3}$$

The effective stiffness is

$$k_e = \frac{(k_1 + k_2) k_3}{k_1 + k_2 + k_3}$$

The equation of motion is $m\ddot{u} + k_e u = p(t)$

(02)

Solution:

In each case the system is equivalent to the spring mass system shown for which the equation of motion is

$$\left(\frac{W}{g}\right)\ddot{u} + Ku = 0$$



The spring stiffness is determined from the deflection u under a vertical force f_s applied at the location of the lumped weight:

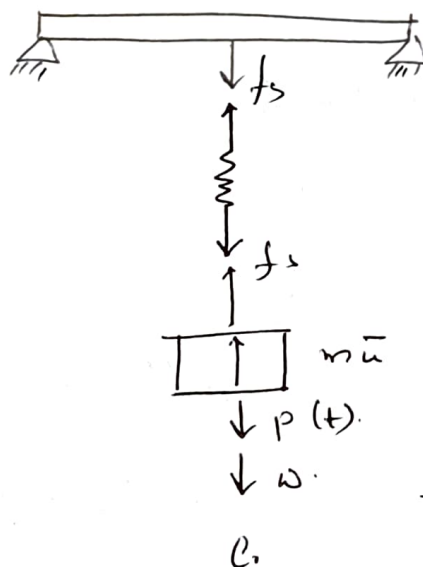
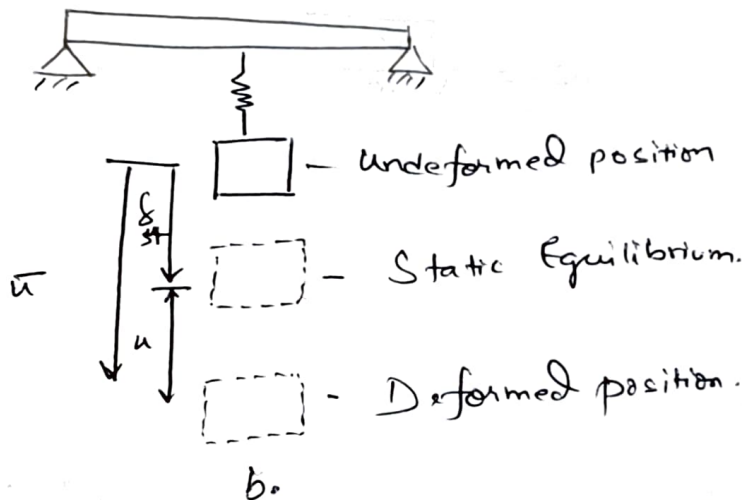
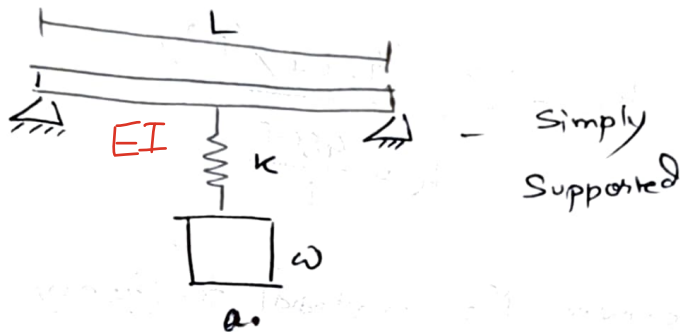
~~Simply supported beam: $u = \frac{f_s l^3}{48EI} \Rightarrow K = \frac{48EI}{l^3}$~~

~~Cantilever beam: $u = \frac{f_s l^3}{3EI} \Rightarrow K = \frac{3EI}{l^3}$~~

Clamped beam: $u = \frac{f_s l^3}{192EI} \Rightarrow K = \frac{192EI}{l^3}$

(03)

Solution +



1. Equation of Motion

$$m\ddot{u} + f_s = \omega + P(t)$$

where,

$$f_s = k_e \bar{u}$$

the equation of motion is:

$$m\ddot{u} + k_e \bar{u} = \omega + P(t)$$

2. Determine the effective stiffness.

$$f_s = k_e \bar{u} \quad \text{--- (1)}$$

where,

$$\bar{u} = \delta_{\text{spring}} + \delta_{\text{beam}}$$

$$f_s = k \cdot \delta_{\text{spring}} = \frac{K_{\text{beam}}}{K_{\text{beam}} + k} \delta_{\text{beam}} \quad \text{--- (2)}$$

Substitute for δ_s from eqn (2) and u from

$$\text{eqn (1)} \quad \frac{f_s}{k_e} = \frac{f_s}{k} + \frac{f_s}{K_{\text{beam}}}$$

$$k_e = \frac{k \cdot k_{beam}}{k + k_{beam}}$$

$$k_e = \frac{k (48EI/L^3)}{k + \frac{48EI}{L^3}}$$

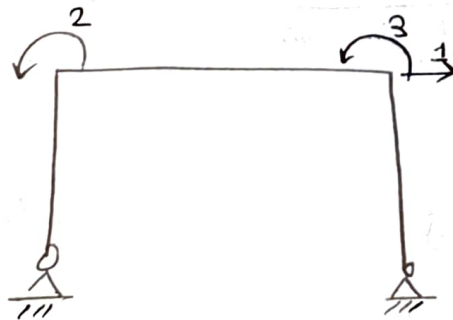
3. Determine the natural frequency

$$\omega_n = \sqrt{\frac{k_e}{m}}$$

(04)

Solution:-

1. Define degrees of freedom (DOF)

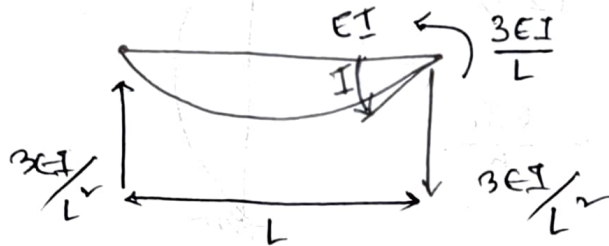


2. Reduced Stiffness Coefficients

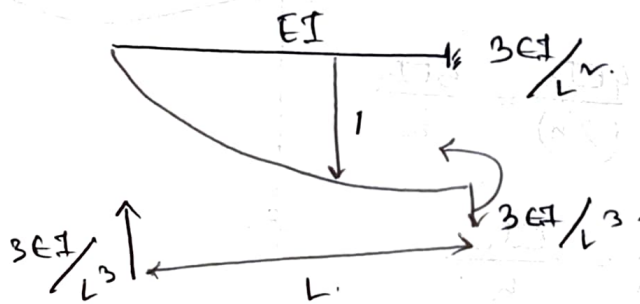
Since there are no external moments applied at the pinned supports the following reduced

Stiffness coefficients are used for the columns.

Joint Rotation



Joint Translation



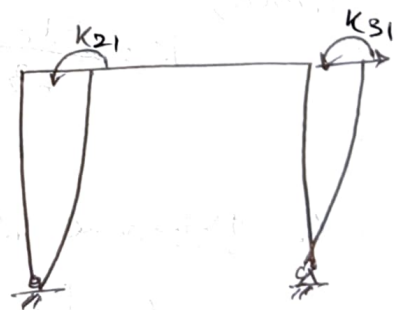
3. Form Structural Stiffness Matrix

$$u_1 = 1, u_2 = u_3 = 0.$$

$$K_{11} = 2 \cdot \frac{3EI_c}{h^3} = \frac{6EI_c}{h^3}.$$

$$K_{21} = K_{31} = \frac{3EI_c}{h^2}.$$

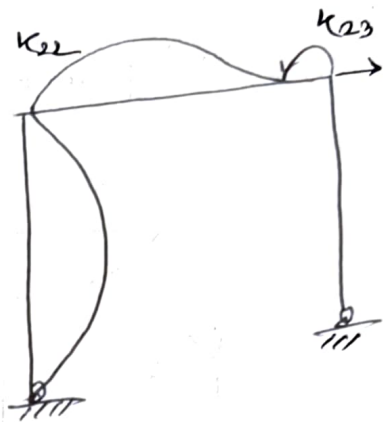
$$u_2 = 1, u_1 = u_3 = 0.$$



$$K_{22} = \frac{3EI_c}{h} + \frac{4EI_c}{(2h)} = \frac{5EI_c}{h}$$

$$K_{32} = \frac{2EI_c}{(2h)} = \frac{EI_c}{h}$$

$$K_{12} = \frac{3EI_c}{h}$$

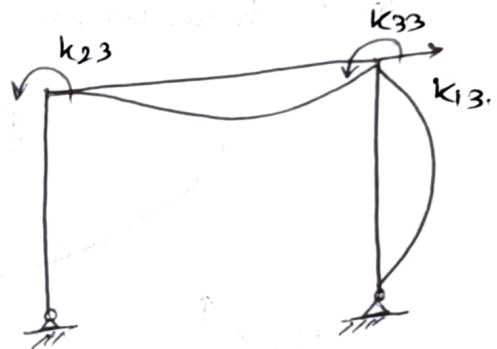


$$u_3 = 1, u_1 = u_2 = 0$$

$$K_{33} = \frac{3EI_c}{h} + \frac{4EI_c}{(2h)} = \frac{5EI_c}{h}$$

$$K_{23} = \frac{2EI_c}{(2h)} = \frac{EI_c}{h}$$

$$K_{13} = \frac{3EI_c}{h}$$



Hence,

$$K = \frac{EI_c}{h^3} \begin{bmatrix} 6 & 3h & 3h \\ 3h & 5h^2 & h^2 \\ 3h & h^2 & 5h^2 \end{bmatrix}$$

4. Determine Lateral Stiffness.

The lateral stiffness k of the frame can be obtained by static condensation since there is no force acting on DOF 2 & 3:

$$\frac{EI_c}{h^3} \begin{bmatrix} 6 & 3h & 3h \\ 3h & 5h^2 & h^2 \\ 3h & h^2 & 5h^2 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \end{Bmatrix} = \begin{Bmatrix} \frac{1}{2} \\ 0 \\ 0 \end{Bmatrix}$$

first partition K as

$$K = \frac{EI_c}{h^3} \begin{bmatrix} 6 & 3h & 3h \\ 3h & 5h^2 & h^2 \\ 3h & h^2 & 5h^2 \end{bmatrix} = \begin{bmatrix} K_{tt} & K_{t0} \\ K_{t0} & K_{00} \end{bmatrix}$$

where, $K_{tt} = \frac{EI_c}{h^3} [6]$

$$K_{t0} = \frac{EI_c}{h^3} [3h \quad 3h]$$

$$K_{00} = \frac{EI_c}{h^3} \begin{bmatrix} 5h^2 & h^2 \\ h^2 & 5h^2 \end{bmatrix}$$

Then compute the lateral stiffness K from.

$$K = K_{tt} - K_{t0} K_{00}^{-1} K_{t0}^T$$

Since, $K_{00}^{-1} = \frac{h}{24EI_c} \begin{bmatrix} 5 & -1 \\ -1 & 5 \end{bmatrix}$

We get,

$$K = \frac{6EI_c}{h^3} - \frac{EI_c}{h^3} [3h \quad 3h] \cdot \frac{h}{24EI_c} \begin{bmatrix} 5 & -1 \\ -1 & 5 \end{bmatrix} \frac{EI_c}{h^3} [3h \\ 3h]$$

$$K = \frac{EI_c}{h^3} [6 - 3]$$

$$K = \frac{3EI_c}{h^3}$$

5. Equation of motion

$$m\ddot{u} + \frac{3EI_c}{h^3} u = p(t)$$

$$K = \frac{3EI_c}{h^3}$$

$$K_{ff} = \frac{EI_c}{h^3}$$

$$K_{dd} = \frac{EI_c}{h^3}$$

$$K_{oo} = \frac{EI_c}{h^3}$$

from combine the lateral stiffness

$$K = K_{ff} + K_{dd} + K_{oo}$$

$$K_{oo} = \frac{1}{\frac{1}{K_{ff}} + \frac{1}{K_{dd}} + \frac{1}{K_{oo}}}$$

one day